

RAUMA KYLMAPIHLAJA, Finland

WMO: 027610

Lat: 61.1447N

Lon: 21.3028E

Elev: 29

StdP: 100.98

Time Zone: 2.00 (EUE)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -16.9	(c) -13.8	(d) -19.3	(e) 0.7	(f) -16.6	(g) -15.9	(h) 0.9	(i) -13.2	(j) 22.2	(k) -0.1	(l) 20.2	(m) 0.3	(n) 6.8	(o) 70	(p) 0.923

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 4.1	(c) 23.6	(d) 20.0	(e) 21.9	(f) 18.7	(g) 20.5	(h) 17.7	(i) 20.6	(j) 23.0	(k) 19.2	(l) 21.2	(m) 18.1	(n) 20.1	(o) 6.6	(p) 110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
19.6	14.4	22.3	18.5	13.4	20.9	17.2	12.3	19.6	59.5	23.3	55.0	21.0	51.4	20.2	23.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 17.8	(b) 16.0	(c) 14.5	(e) -16.3	(f) 25.9	(g) 5.6	(h) 3.0	(i) -20.3	(j) 28.1	(k) -23.6	(l) 29.8	(m) -26.7	(n) 31.5	(o) -30.7	(p) 33.7		
(a) 17.8	(b) 16.0	(c) 14.5	(e) -17.2	(f) 20.8	(g) 5.3	(h) 1.8	(i) -21.0	(j) 22.1	(k) -24.2	(l) 23.1	(m) -27.2	(n) 24.1	(o) -31.1	(p) 25.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	6.3	-3.4	-3.0	-0.8	3.2	9.0	13.1	16.9	16.8	13.0	7.3	3.3	0.0
	DBStd	7.95	4.66	5.05	3.33	2.46	3.34	2.94	2.94	2.43	2.55	2.89	3.68	4.73
	HDD10.0	1987	416	363	334	206	58	5	0	0	6	89	201	309
	HDD18.3	4410	674	596	593	454	289	161	64	59	159	341	451	568
	CDD10.0	653	0	0	0	1	28	97	213	211	97	6	0	0
	CDD18.3	34	0	0	0	0	0	3	19	12	1	0	0	0
	CDH23.3	67	0	0	0	0	1	6	31	29	0	0	0	0
	CDH26.7	7	0	0	0	0	0	0	2	4	0	0	0	0
	WSAvg	7.9	8.4	8.2	8.2	7.2	7.1	7.0	6.8	7.2	7.7	8.7	8.5	9.3
	PrecAvg	630	51	37	33	32	38	56	68	71	58	69	60	58
(15) Precipitation	PrecMax	818	100	78	58	61	85	108	134	137	128	159	139	108
	PrecMin	514	7	2	5	1	5	18	2	25	12	7	9	7
	PrecStd	77	23	20	13	16	20	23	33	32	35	35	28	26
	PrecStd	77	23	20	13	16	20	23	33	32	35	35	28	26
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	3.7	3.6	6.8	13.8	20.8	24.2	26.1	26.8	20.6	13.9	10.1	7.5
		MCWB	2.7	2.5	3.2	9.4	14.9	17.2	21.3	21.1	16.8	12.2	9.2	6.6
	2%	DB	3.1	2.9	4.5	9.7	17.4	20.7	23.9	23.3	18.2	12.7	9.3	6.3
		MCWB	2.3	2.1	3.0	6.9	13.5	16.8	20.6	19.8	15.8	11.3	8.7	5.4
	5%	DB	2.4	2.4	3.5	7.7	15.3	18.8	22.4	21.4	17.1	11.7	8.4	5.3
		MCWB	1.7	1.7	2.4	5.6	12.5	16.0	19.7	18.2	15.1	10.5	7.7	4.4
	10%	DB	1.8	1.9	2.7	6.4	13.4	17.1	21.0	19.9	16.2	10.9	7.5	4.6
		MCWB	1.2	1.3	1.8	4.7	11.1	14.6	18.7	17.2	14.5	9.8	6.7	3.7
	0.4%	WB	3.3	2.9	4.1	9.4	15.7	19.0	22.1	21.9	17.2	12.6	9.5	6.9
		MCDB	3.5	3.2	5.5	13.2	19.5	21.5	24.8	25.2	19.5	13.3	9.8	7.3
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	2.5	2.3	3.2	7.2	14.2	17.2	21.1	20.1	16.2	11.7	8.9	5.5
		MCDB	2.8	2.6	4.1	9.2	16.9	19.9	23.4	22.7	17.8	12.6	9.3	6.1
	5%	WB	1.8	1.8	2.6	6.0	12.8	16.1	20.1	18.7	15.5	10.9	7.9	4.6
		MCDB	2.3	2.2	3.3	7.4	14.9	18.7	22.1	20.9	16.9	11.5	8.3	5.2
	10%	WB	1.2	1.4	2.0	5.0	11.3	14.9	19.0	17.7	14.8	10.0	6.8	3.8
		MCDB	1.7	1.8	2.6	6.1	13.3	16.9	20.5	19.6	16.1	10.7	7.4	4.4
(35) Mean Daily Temperature Range	5% DB	MDBR	3.9	3.5	3.9	3.9	4.5	4.1	4.1	3.8	3.4	2.8	2.6	2.9
		MCDBR	2.5	2.3	3.5	6.0	7.5	6.9	6.0	5.6	4.4	3.1	2.5	2.6
		MCWBR	2.4	2.0	2.6	3.9	4.2	3.9	3.4	3.4	3.0	2.7	2.4	2.8
		MCDBR	2.4	2.4	3.2	5.7	7.5	6.3	5.3	4.7	4.1	2.9	2.6	2.6
	5% WB	MCWBR	2.4	2.2	2.6	3.9	4.5	3.8	3.3	3.3	2.9	2.9	2.6	2.8
		MCWBR	2.4	2.2	2.6	3.9	4.5	3.8	3.3	3.3	2.9	2.9	2.6	2.8
(40) Clear-Sky Solar Irradiance	taub		0.268	0.310	0.327	0.367	0.365	0.356	0.383	0.370	0.345	0.328	0.288	0.361
	taud		2.152	2.249	2.320	2.327	2.417	2.478	2.427	2.451	2.503	2.401	2.196	2.560
	Ebn at Noon		454	663	791	825	858	874	835	817	776	644	419	301
	Edh at Noon		38	66	86	103	102	98	101	91	73	57	36	25
(44) All-Sky Solar Radiation	RadAvg		0.20	0.78	2.32	3.96	5.54	6.09	5.62	4.37	2.61	1.08	0.28	0.12
	RadStd		0.03	0.15	0.26	0.39	0.45	0.37	0.49	0.38	0.27	0.17	0.05	0.01

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (46 neighbors)	+0.42	N/A	N/A	N/A	N/A	N/A	N/A	-118	N/A	N/A	N/A

Nomenclature: See separate page